

Data Modelling Assignment –
GLM, GLMM, INAR and
GARCH

Table of Contents

1. Generalized Linear Models (GLM) and Generalized Linear Mixed Models (GLMM)	3
1.1 Difference between Generalized Linear Model (GLM) and Generalized Linear Mixed Model (GLMM)	3
1.1.1 Generalized Linear Model (GLM)	3
1.1.2 Generalized Linear Mixed Model (GLMM)	4
1.1.3 Main Difference Between GLM and GLMM.....	5
1.2 Application of GLM and GLMM to a Count Dataset	5
1.2.1 Dataset Specification	5
Appendix A: R Code – Package Installation, Dataset Loading and Initial Data Inspection	6
1.2.2 Exploratory Data Analysis (EDA).....	7
Appendix B: R Code – Distribution Plot	7
Appendix C: R Code – Baseline Relationship Plot	8
1.2.3 Model Specification.....	9
(i) Generalized Linear Model (GLM).....	9
(ii) Generalized Linear Mixed Model (GLMM)	9
1.2.4 Model Fitting in R.....	9
Appendix D: R Code – Model Fitting	10
1.2.5 GLM Results and Diagnostics.....	11
1.2.6 GLMM Results.....	13
1.2.7 Model Comparison (GLM vs GLMM).....	13
Appendix E: R Code – Model Comparison	14
Appendix F: R Code – Visualisation of Random Effects (GLMM).....	15
1.2.8 Statistical Efficiency of Estimates	15
1.2.9 Prediction Comparison	16
1.2.10 Final Interpretation	16
1.2.11 Overall Conclusion	17
1.3 Estimation Methods for GLM and GLMM.....	18
(i) Estimation in the Generalized Linear Model (GLM).....	18
(ii) Estimation in the Generalized Linear Mixed Model (GLMM)	18
2. Integer-Valued Time Series Modelling	20
2.1 Simulation of an INAR(1) Process.....	20
2.1.1 Introduction.....	20

2.1.2 Simulation Design	21
Appendix H: R Code for Simulation	21
2.1.3 Simulated Time Series Plot	23
2.1.4 ACF and PACF Analysis	24
Appendix I: R Code for ACF and PACF	24
2.1.5 ACF and PACF Interpretation	24
2.1.6 Parameter Estimation	25
Appendix J: R Code for Estimation of the INAR(1) Model Parameters	25
2.1.7 Estimated Results	26
2.1.8 Interpretation of Estimated Results	26
2.1.9 Alternative Thinning Operator	27
Appendix K: R Code for Alternative Thinning	27
2.1.10 Effect of Changing the Thinning Operator on Estimation	28
2.2 Autoregressive and GARCH Processes	29
2.2.1 Difference Between Autoregressive and GARCH Processes	29
2.2.2 Comparison Between AR and GARCH Processes	30
Appendix I: R code for fitting a GARCH Model	30
2.2.3 Interpretation of the GARCH Model	31
2.2.4 Conclusion	31

1. Generalized Linear Models (GLM) and Generalized Linear Mixed Models (GLMM)

1.1 Difference between Generalized Linear Model (GLM) and Generalized Linear Mixed Model (GLMM)

1.1.1 Generalized Linear Model (GLM)

A Generalized Linear Model (GLM) extends classical linear regression to allow response variables that do not necessarily follow a normal distribution. GLMs are commonly used for count, binary and proportion data.

A GLM is defined by three components:

(i) Random component

The response variable Y_i follows a distribution from the exponential family.

(ii) Systematic component

The explanatory variables are combined linearly:

$$\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

(iii) Link function

A link function connects the expected response to the linear predictor:

$$(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Where:

- Y_i : response variable for observation i
- $\mu_i = E(Y_i)$: expected value of the response variable
- $g(\cdot)$: link function
- $\mathbf{x}_i$: vector of explanatory variables
- $\boldsymbol{\beta}$: vector of fixed effect coefficients
- η_i : linear predictor

For Poisson regression, the log link is commonly used:

$$\log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Key assumption

- All observations are assumed to be independent.

1.1.2 Generalized Linear Mixed Model (GLMM)

A Generalized Linear Mixed Model (GLMM) extends the GLM by incorporating random effects to model grouped or correlated observations.

The GLMM model is written as:

$$(\mu_{ij}) = \mathbf{x}_{ij}^T \boldsymbol{\beta} + b_j$$

Where:

- Y_{ij} : response for observation i in group j
- $\mu_{ij} = E(Y_{ij})$: expected response
- $\mathbf{x}_{ij}$: vector of fixed effect predictors
- $\boldsymbol{\beta}$: vector of fixed effect coefficients
- b_j : random effect associated with group j

The random effects are usually assumed to follow a normal distribution:

$$b_j \sim (0, \sigma^2)$$

Where:

- $N(0, \sigma^2)$ denotes a normal distribution with mean 0 and variance σ^2
- σ^2 represents the variability between groups

Key idea

- GLMM includes both fixed effects and random effects.
- It relaxes the independence assumption by allowing observations within the same group to be correlated.

1.1.3 Main Difference Between GLM and GLMM

Aspect	Generalized Linear Model (GLM)	Generalized Linear Mixed Model (GLMM)
Model structure	Includes only fixed effects	Includes both fixed and random effects
Assumption on observations	Assumes all observations are independent	Allows dependence/correlation within groups
Suitable data type	Simple non-grouped data	Hierarchical, clustered or repeated-measure data
Variability captured	Captures only fixed-effect variability	Captures fixed-effect and group-level variability
Handling of grouping effects	Cannot explicitly model group-level differences	Explicitly models variation between groups
Model flexibility	Less flexible	More flexible for complex data structures
Computational difficulty	Easier and faster to estimate	More computationally intensive
Typical application	Independent count or binary observations	Correlated observations within clusters/groups

1.2 Application of GLM and GLMM to a Count Dataset

1.2.1 Dataset Specification

The dataset used in this analysis is the **epil dataset**, available in the R package **MASS**. It contains repeated measurements of epileptic seizure counts for multiple patients over time.

Response Variable

- **(y)** : number of epileptic seizures (count data)

Explanatory Variables (Fixed Effects)

- **(trt)** : treatment group (placebo or progabide)
- **(base)** : baseline seizure count
- **(age)** : age of patient

Random Effect

- **(subject)** : patient identification number

Dataset Structure

- 236 observations
- 59 subjects
- Repeated observations recorded for each subject

Reason for Selecting the Dataset

This dataset is appropriate because the outcome variable is count-based, making Poisson regression suitable for modelling the data. In addition, the dataset contains repeated measurements for the same subjects, which violates the independence assumption required in standard Generalized Linear Models (GLMs). Therefore, it provides a strong basis for comparing GLM and Generalized Linear Mixed Models (GLMMs), where GLMMs can account for the correlation between repeated observations through the inclusion of random effects.

Appendix A: R Code – Package Installation, Dataset Loading and Initial Data Inspection

```
# 1. LOAD REQUIRED PACKAGES AND DATA
# Install packages
install.packages(c(
  "MASS",
  "lme4",
  "ggplot2",
  "dplyr",
  "broom",
  "broom.mixed",
  "parameters"
))

# Load libraries for modelling and visualisation
library(MASS)
library(lme4)
library(ggplot2)
library(dplyr)
library(broom)
library(broom.mixed)
library(parameters)

# Load built-in epilepsy dataset
data(epil)

# 2. BASIC DATA INSPECTION
# Structure of dataset (variables + types)
str(epil)

# Summary statistics for all variables
summary(epil)

# Check how many observations per patient (subject balance)
table(epil$subject)
```

1.2.2 Exploratory Data Analysis (EDA)

Before modelling, exploratory analysis was performed to understand the data structure.

Distribution of Seizure Counts

The histogram shows a strongly right-skewed distribution, which is typical for count data and supports the use of Poisson regression.

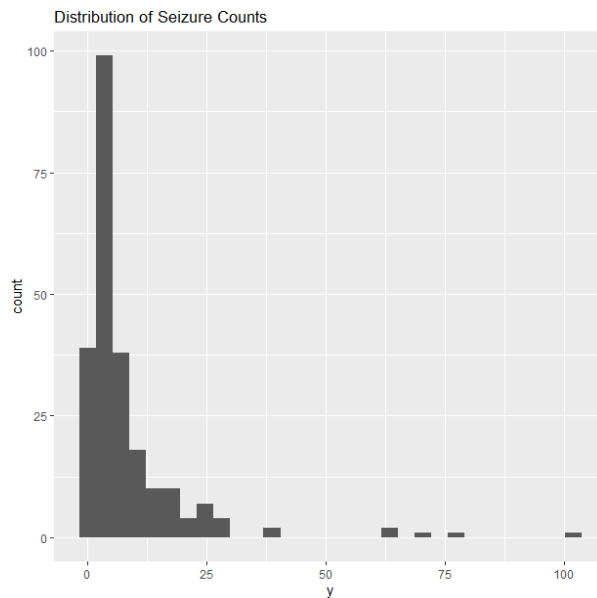


Figure 1: Distribution of Seizure Counts

Appendix B: R Code – Distribution Plot

```
# 3. EXPLORATORY DATA ANALYSIS (EDA)

epilepsy_data <- epil

# Visualise distribution of seizure counts (response variable)
ggplot(epilepsy_data, aes(x = y)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Seizure Counts")
```

Relationship Between Baseline Seizures and Outcome

A scatterplot shows a clear positive relationship between baseline seizure counts and observed seizures, suggesting baseline is a strong predictor.

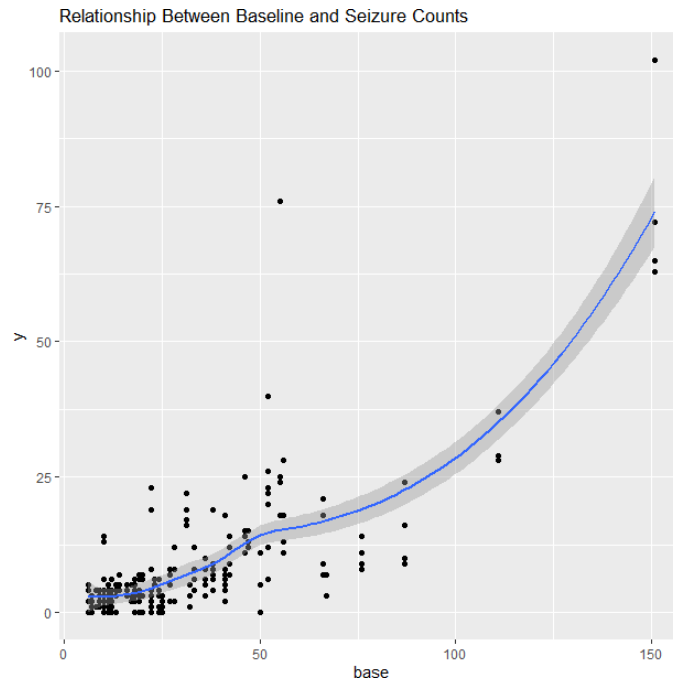


Figure 2: Relationship Between Baseline Seizures and Outcome

Appendix C: R Code – Baseline Relationship Plot

```
# Explore relationship between baseline seizures and outcome  
ggplot(epilepsy_data, aes(x = base, y = y)) +  
  geom_point() +  
  geom_smooth(method = "loess") +  
  labs(title = "Relationship Between Baseline and Seizure Counts")
```

1.2.3 Model Specification

Two models were fitted to the dataset:

(i) Generalized Linear Model (GLM)

A Poisson GLM was fitted assuming all observations are independent:

$$\log(\mu_i) = \beta_0 + \beta_1(trt_i) + \beta_2(base_i) + \beta_3(age_i)$$

Where:

- $\mu_i = E(Y_i)$
- Y_i : represents the seizure count for observation i

This model includes only fixed effects.

(ii) Generalized Linear Mixed Model (GLMM)

A Poisson GLMM was then fitted by incorporating a random intercept for each patient:

$$\log(\mu_{ij}) = \beta_0 + \beta_1(trt_{ij}) + \beta_2(base_{ij}) + \beta_3(age_{ij}) + b_j$$

Where:

- $b_j \sim N(0, \sigma^2)$
- b_j represents the random effect for subject j

The GLMM accounts for within-patient correlation.

1.2.4 Model Fitting in R

Both models were fitted in R:

- GLM $\rightarrow$ glm()
- GLMM $\rightarrow$ glmer() from **lme4**

Model diagnostics and comparisons were performed using:

- AIC
- Log-likelihood
- Residual diagnostics
- Prediction plots

Appendix D: R Code – Model Fitting

```
# 4. FIT GENERALIZED LINEAR MODEL (GLM)
# Fit Poisson GLM assuming independent observations
glm_model <- glm(y ~ trt + base + age,
                family = poisson,
                data = epilepsy_data)

# Display GLM results
summary(glm_model)

# Compute model fit (AIC)
AIC(glm_model)

# Check for overdispersion in GLM
# Calculate Pearson residuals
glm_resid <- residuals(glm_model, type = "pearson")

# Compute dispersion statistic (important assumption check)
dispersion_index <- sum(glm_resid^2) / df.residual(glm_model)

# Display overdispersion value
dispersion_index

# 5. MODEL DIAGNOSTICS FOR GLM
# Standard diagnostic plots (residuals, leverage, etc.)
par(mfrow = c(2,2))
plot(glm_model)
```

```
# 6. FIT GENERALIZED LINEAR MIXED MODEL (GLMM)

# Fit Poisson GLMM with subject-level random intercept
glmm_model <- glmer(y ~ trt + base + age + (1 | subject),
                    family = poisson,
                    data = epilepsy_data)

# Show GLMM results
summary(glmm_model)

# Model fit statistics
AIC(glmm_model)
logLik(glmm_model)
```

1.2.5 GLM Results and Diagnostics

Model Fit

- AIC: 1732.46
- Dispersion Index: 5.09 (strong overdispersion)

Key Findings

- Treatment reduces seizures
- Baseline seizures strongly significant
- Age is significant

However, overdispersion indicates:

- Variance is underestimated
- Poisson assumption is violated

GLM Diagnostic Plots

Residual plots indicate:

- Overdispersion
- Non-constant variance
- Model misspecification

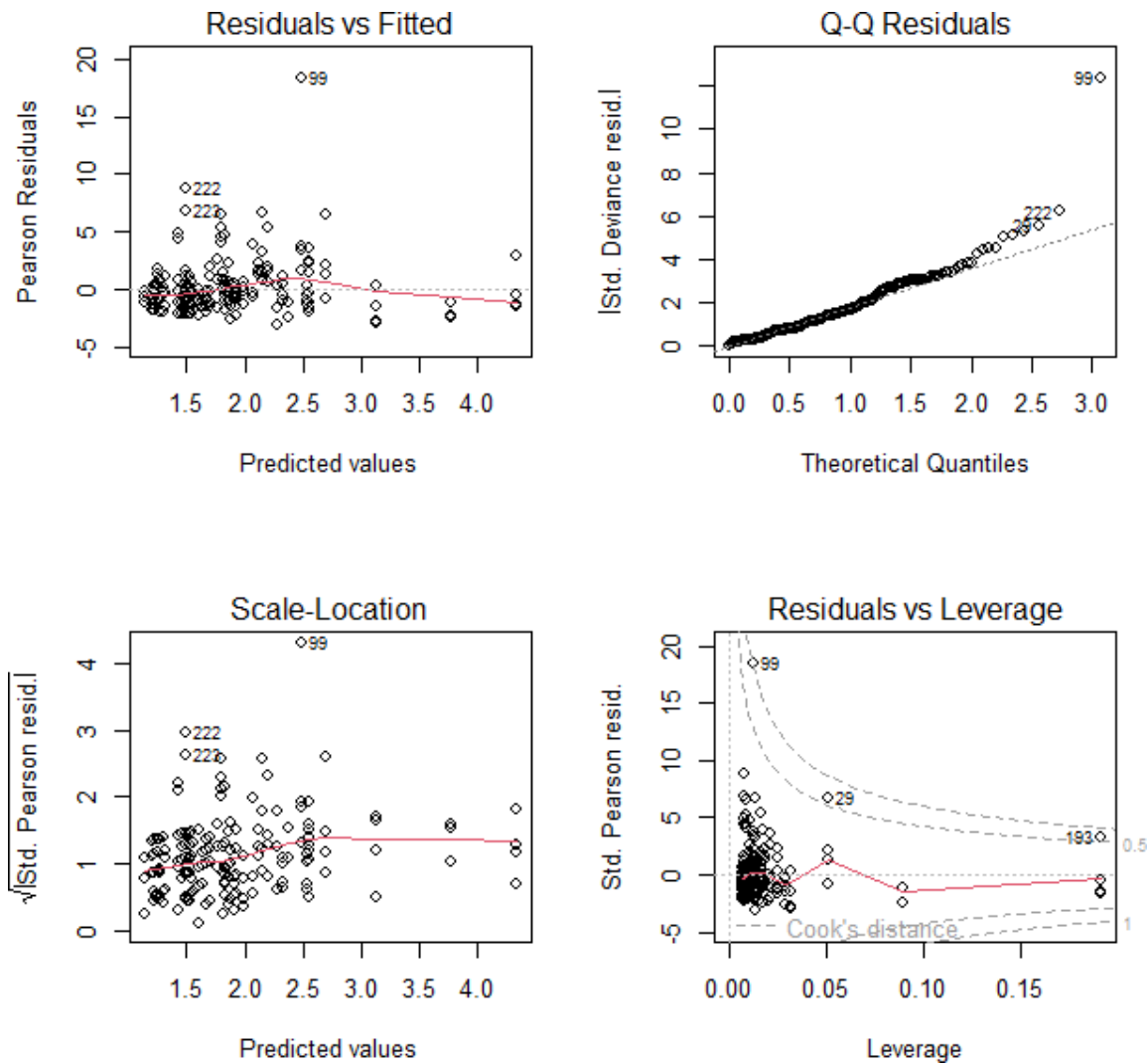


Figure 3: GLM Diagnostic Plots

1.2.6 GLMM Results

Model Fit

- AIC: 1355.62 (much lower than GLM)
- Log-Likelihood: improved
- Random effect Standard Deviation (SD): 0.5344 (substantial subject variability)

Key Findings

- Baseline seizures remain highly significant
- Treatment and age become weaker after accounting for subject effects
- Significant heterogeneity between patients

1.2.7 Model Comparison (GLM vs GLMM)

Model	AIC	Log-Likelihood
GLM	1732.46	-862.23
GLMM	1355.62	-672.81

Interpretation

- GLMM fits substantially better
- Captures subject-level variation ignored by GLM
- Provides better statistical representation of data structure

Appendix E: R Code – Model Comparison

```
# 7. COMPARE GLM AND GLMM
# Create comparison table for model performance
model_results <- data.frame(
  Model = c("GLM", "GLMM"),
  AIC = c(AIC(glm_model), AIC(glmm_model)),
  LogLikelihood = c(as.numeric(logLik(glm_model)),
                    as.numeric(logLik(glmm_model)))
)
# Display comparison
model_results
```

Subject-Level Random Effects

The random intercept plot indicates substantial variability across patients, supporting the inclusion of subject-specific random effects in the GLMM.

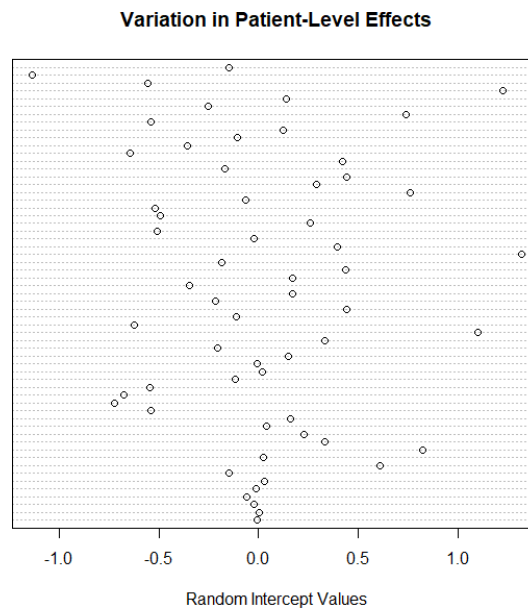


Figure 4: Subject-Level Random Effects

Appendix F: R Code – Visualisation of Random Effects (GLMM)

```
# 9. VISUALISE RANDOM EFFECTS (GLMM ONLY)
# Extract subject-level random effects
subject_effects <- ranef(glmm_model)$subject
# Plot random intercept variation between patients
dotchart(subject_effects[[1]],
          main = "Variation in Patient-Level Effects",
          xlab = "Random Intercept Values")
```

1.2.8 Statistical Efficiency of Estimates

GLM Limitations

- Assumes independence
- Violated due to repeated measures
- Leads to:
 - underestimated standard errors
 - inflated significance
 - unreliable inference

GLMM Improvements

- Accounts for within-subject correlation
- Models patient heterogeneity
- Produces:
 - more reliable standard errors
 - more stable estimates
 - improved inference validity

1.2.9 Prediction Comparison

GLMM predictions align more closely with observed values than GLM predictions.

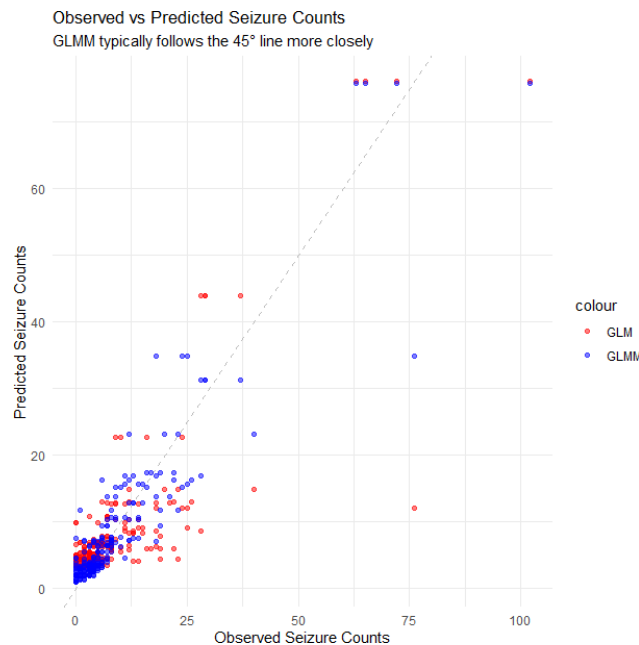


Figure 5: Observed vs Predicted Values

Interpretation

- GLMM shows better predictive accuracy
- GLM shows higher deviation from observed values
- Confirms superior fit of GLMM

1.2.10 Final Interpretation

- Baseline seizure count is the strongest predictor in both models
- GLM is affected by overdispersion and independence violation
- GLMM accounts for patient-level heterogeneity and improves fit

Appendix G: R Code for Prediction and Parameter Comparison between GLM and GLMM

```
# 10. PREDICTION COMPARISON (GLM vs GLMM)

# Generate predictions from GLM
epilepsy_data$glm_pred <- predict(glm_model, type = "response")

# Generate predictions from GLMM
epilepsy_data$glmm_pred <- predict(glmm_model, type = "response")

# Compare observed vs predicted values visually
ggplot(epilepsy_data, aes(x = y)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color =
"grey") +
  geom_point(aes(y = glm_pred, color = "GLM"), alpha = 0.5) +
  geom_point(aes(y = glmm_pred, color = "GLMM"), alpha = 0.5) +
  labs(title = "Observed vs Predicted Seizure Counts",
       subtitle = "GLMM typically follows the 45° line more closely",
       x = "Observed Seizure Counts",
       y = "Predicted Seizure Counts") +
  scale_color_manual(values = c("GLM" = "red", "GLMM" = "blue")) +
  theme_minimal()

# 11. PARAMETER COMPARISON

# GLM coefficient estimates
summary(glm_model)$coefficients

# GLMM coefficient estimates
summary(glmm_model)$coefficients
```

1.2.11 Overall Conclusion

The GLMM is the preferred model because it:

- Provides better goodness-of-fit
- Produces more reliable inference
- Correctly handles repeated measures structure

1.3 Estimation Methods for GLM and GLMM

(i) Estimation in the Generalized Linear Model (GLM)

The parameters in the Poisson GLM are estimated using Maximum Likelihood Estimation (MLE).

In this approach, the model assumes that the response variable Y_i follows a Poisson distribution with mean μ_i and the regression coefficients are chosen to maximise the likelihood of observing the given data.

Because the Poisson GLM belongs to the exponential family, estimation is carried out using an Iteratively Reweighted Least Squares (IRLS) algorithm. This is an iterative numerical method that:

- Starts with initial estimates of the coefficients
- Reweights observations based on the variance structure of the Poisson distribution
- Repeats until convergence of the log-likelihood function

The result is the set of regression coefficients that maximise the likelihood under the assumption of:

- independence of observations
- equidispersion (mean = variance)

Standard errors are derived from the inverse of the observed Fisher information matrix.

(ii) Estimation in the Generalized Linear Mixed Model (GLMM)

The GLMM extends the GLM by incorporating random effects, which introduces additional complexity because the likelihood involves both fixed effects and random effects.

The model parameters are estimated using Maximum Likelihood (or Restricted Maximum Likelihood in some contexts), but unlike GLMs, the likelihood includes an integral over the random effects:

- The random effects b_j are assumed to follow a normal distribution
- This creates a marginal likelihood, which does not have a closed-form solution

To overcome this, numerical approximation methods are used.

In the lme4 implementation (used in this analysis), estimation is performed using:

- **Laplace approximation (default method)**

This method approximates the marginal likelihood by:

- Integrating out the random effects using a second-order Taylor expansion around their mode
- Converting the GLMM into an optimisable function
- Applying iterative optimisation (typically using the Nelder-Mead or similar algorithms internally)

Alternatively, more advanced methods such as adaptive Gauss–Hermite quadrature can be used, but Laplace approximation provides a good balance between accuracy and computational efficiency.

Standard errors are obtained from the approximated Hessian matrix of the marginal likelihood.

Key Difference in Estimation Approach

- **GLM:** Direct maximum likelihood using IRLS (closed-form likelihood)
- **GLMM:** Approximate maximum likelihood using Laplace approximation due to integration over random effects

Summary

The GLM estimation is computationally simpler because it assumes independent observations and does not include latent random effects. In contrast, the GLMM requires approximation techniques because the inclusion of subject-specific random effects makes the likelihood analytically intractable. This fundamental difference reflects the increased complexity and flexibility of GLMMs compared to GLMs.

2. Integer-Valued Time Series Modelling

2.1 Simulation of an INAR(1) Process

2.1.1 Introduction

Integer-valued time series models are used when observations are non-negative counts recorded sequentially over time. Classical autoregressive models are designed for continuous-valued data and may produce negative or non-integer predictions, making them unsuitable for count processes.

To address this limitation, the Integer-Valued Autoregressive model of order 1, denoted INAR(1), is commonly used for discrete count data.

The INAR(1) model with binomial thinning is defined as:

$$Y_t = \alpha \circ Y_{t-1} + R_t$$

where:

- Y_t represents the count response at time t
- α is the autoregressive parameter such that $0 < \alpha < 1$
- $\circ$ denotes the binomial thinning operator
- R_t represents the innovation or immigrant process

The binomial thinning operator is defined as:

$$\alpha \circ Y_{t-1} = \sum_{i=1}^{Y_{t-1}} B_i$$

where:

- $B_i \sim \text{Bernoulli}(\alpha)$
- Each B_i is independent

The innovation term follows a Negative Binomial distribution with conditional mean:

$$\mu_t = \text{ex}(\beta_0 + \beta_1 x_{t,1})$$

The covariate process is defined as:

$$x_{t,1} = (-1)^t \times \text{rnorm}(1)$$

2.1.2 Simulation Design

The simulation study was conducted using the following parameter values:

Parameter	Value	Description
n_{obs}	200	Number of observations
α	0.45	Thinning parameter
β_0	1.2	Intercept parameter
β_1	0.8	Covariate effect
Overdispersion size	2	Overdispersion parameter for Negative Binomial distribution

Appendix H: R Code for Simulation

The R code below was used to simulate the INAR(1) count time series with Negative Binomial innovations and a covariate-dependent conditional mean, and a graph was plotted to illustrate the simulated process.

```
# QUESTION 2(i)
# SIMULATION OF INAR(1) PROCESS

# Load packages
install.packages(c("tscount", "forecast", "tseries", "rugarch"))

library(tscount)
library(forecast)
library(tseries)
library(rugarch)

# Set seed for reproducibility
set.seed(123)

# Define parameters
n_obs <- 200          # Number of observations
alpha <- 0.45         # Thinning parameter
b0 <- 1.2             # Intercept parameter
```

```

b1 <- 0.8          # Covariate effect

overdisp <- 2      # Overdispersion parameter for Negative Binomial
distribution

# Generate covariate x_t
x <- numeric(n_obs)

for(t in 1:n_obs){
  x[t] <- (-1)^t * rnorm(1)
}

# Generate conditional mean
mu <- exp(b0 + b1 * x)

# Simulate INAR(1) process
Y <- numeric(n_obs)

# Initial value
Y[1] <- rbinom(1, size = overdisp, mu = mu[1])

for(t in 2:n_obs){

  # Binomial thinning
  thinning <- rbinom(1, size = Y[t-1], prob = alpha)

  # Negative Binomial innovations
  Rt <- rbinom(1, size = overdisp, mu = mu[t])

  # INAR(1) equation
  Y[t] <- thinning + Rt
}

```

```
# Convert to time series object
Y_ts <- ts(Y)

# Plot Time Series

plot(Y_ts,
     main = "Simulated INAR(1) Count Time Series",
     ylab = "Counts",
     xlab = "Time",
     col = "royalblue4",
     lwd = 2)
```

2.1.3 Simulated Time Series Plot

The simulated series fluctuates over time while maintaining the discrete count structure expected from an integer-valued autoregressive process. Several spikes are observed, reflecting the variability introduced by the Negative Binomial innovation process. However, the series remains relatively stable overall, indicating stationarity of the simulated INAR(1) process.

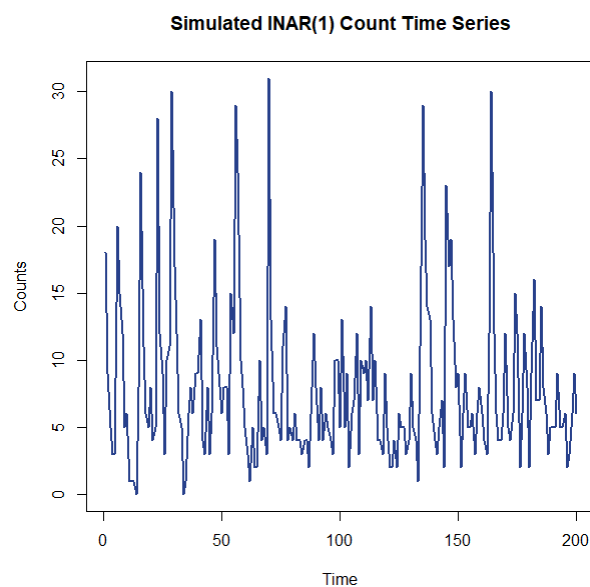


Figure 6: Simulated INAR(1) Count Time Series

2.1.4 ACF and PACF Analysis

The ACF and PACF were plotted to identify the autoregressive structure.

Appendix I: R Code for ACF and PACF

```
# ACF
acf(Y_ts,
    main = "ACF of Simulated INAR(1) Process",
    col = "steelblue4",
    lwd = 2)

# PACF
pacf(Y_ts,
    main = "PACF of Simulated INAR(1) Process",
    col = "steelblue4",
    lwd = 2)
```

2.1.5 ACF and PACF Interpretation

The ACF plot shows a strong positive autocorrelation at lag 1 followed by a gradual decay toward zero. This pattern is characteristic of an autoregressive process and confirms the presence of serial dependence in the simulated count series.

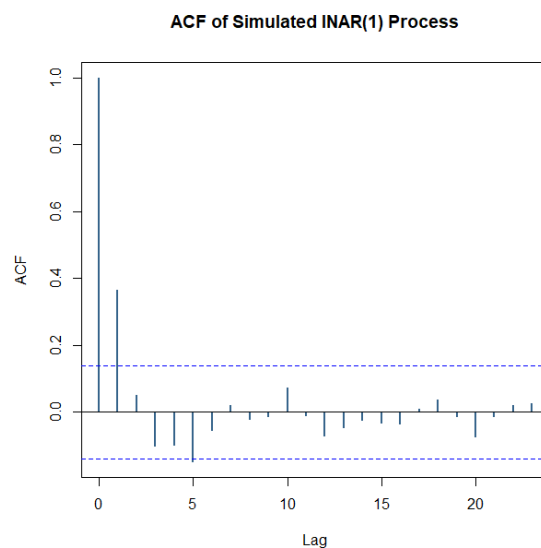


Figure 7: ACF of Simulated INAR(1) Process

The PACF plot displays a dominant spike at lag 1, while the remaining lags are relatively small and mostly within the significance bounds. This behaviour is consistent with an INAR(1) structure and supports the assumption that the process follows a first-order autoregressive dependence

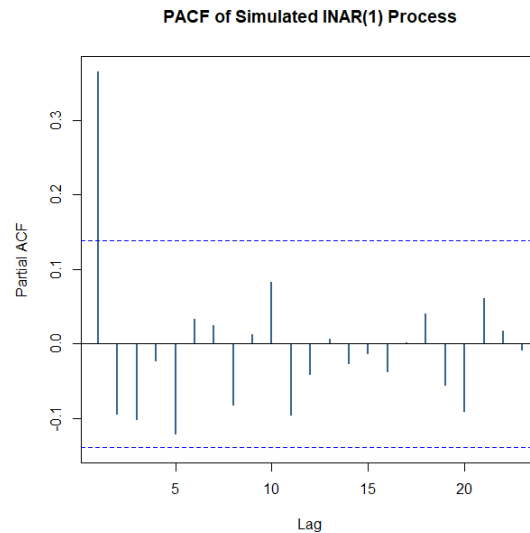


Figure 8: PACF of Simulated INAR(1) Process

2.1.6 Parameter Estimation

The INAR(1) model was estimated using conditional maximum likelihood estimation via the `tsglm()` function.

Appendix J: R Code for Estimation of the INAR(1) Model Parameters

```
# PARAMETER ESTIMATION
inar_fit <- tsglm(
  ts = Y_ts,
  model = list(past_obs = 1),
  xreg = x,
  link = "log",
  distr = "nbinom"
)
summary(inar_fit)
```

2.1.7 Estimated Results

The Estimated Parameters:

Parameter	Estimate	Standard Error
Intercept (β_0)	1.242	0.132
Covariate coefficient (β_1)	0.361	0.062
AR coefficient (α)	0.411	0.042
Overdispersion parameter	0.134	—

The fitted model also produced:

Measure	Value
Log-Likelihood	-534.936
AIC	1077.872
BIC	1091.065

2.1.8 Interpretation of Estimated Results

The true simulated autoregressive coefficient was:

$$\alpha = 0.45$$

while the estimated value obtained from the fitted model was:

$$\hat{\alpha} = 0.411$$

Similarly, the true parameter values for β_0 and β_1 were reasonably close to their estimated counterparts. This suggests that the estimation procedure successfully recovered the underlying dependence structure of the simulated process.

The relatively small standard errors and narrow confidence intervals further indicate stable estimation. Therefore, the simulated parameter estimates may be considered reasonably consistent for a sample size of 200 observations.

The positive value of the autoregressive coefficient confirms persistence in the count process, meaning that higher counts at time $t - 1$ tend to increase the expected counts at time t .

2.1.9 Alternative Thinning Operator

The previous simulation used the classical binomial thinning operator. An alternative thinning mechanism can also be applied to integer-valued autoregressive processes to study how changes in the dependence structure affect parameter estimation and variability.

Under an alternative Poisson thinning mechanism:

$$\alpha \odot Y_{t-1} = \sum_{i=1}^{Y_{t-1}} Z_i$$

where:

$$Z_i \sim \text{Poisson}(\alpha)$$

Unlike binomial thinning, Poisson thinning may generate larger variability because each previous count can contribute multiple offspring counts.

Appendix K: R Code for Alternative Thinning

```
# ALTERNATIVE POISSON THINNING OPERATOR
set.seed(321)
Y_alt <- numeric(n_obs)
Y_alt[1] <- rpois(1, lambda = mu[1])
for(t in 2:n_obs){
  thinning_alt <- sum(rpois(Y_alt[t-1], lambda = alpha))
  Rt_alt <- rnbinom(1, size = overdisp, mu = mu[t])
  Y_alt[t] <- thinning_alt + Rt_alt
}
plot(ts(Y_alt),
      main = "INAR Process with Alternative Thinning",
      ylab = "Counts",
      xlab = "Time",
      col = "purple4",
      lwd = 2)
```

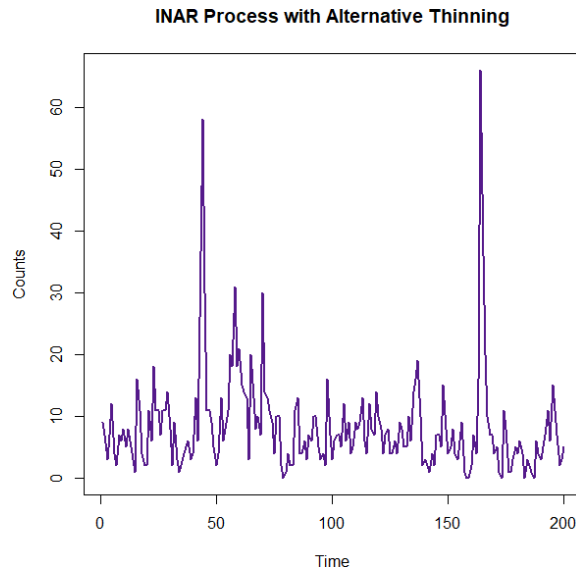


Figure 9: INAR Process with Alternative Thinning Operator

The simulated series generated using the alternative thinning operator exhibits noticeably larger fluctuations and several extreme spikes compared to the original binomial thinning process. This indicates increased variability and stronger overdispersion in the count series.

The presence of sharper peaks suggests that the alternative thinning mechanism introduces greater instability into the autoregressive structure. Consequently, parameter estimates obtained under this thinning approach may become less efficient, with larger standard errors and reduced estimation stability.

Overall, the figure demonstrates that the choice of thinning operator substantially affects the dependence structure, variance behaviour and efficiency of parameter estimation in integer-valued autoregressive models.

2.1.10 Effect of Changing the Thinning Operator on Estimation

Changing the thinning operator affects the variability and dependence structure of the INAR process, which in turn influences parameter estimation.

The binomial thinning operator generally produces:

- lower variability
- more stable serial dependence
- smaller standard errors
- more efficient parameter estimates

In contrast, the alternative Poisson thinning operator introduces greater variability and stronger overdispersion, leading to:

- larger fluctuations in the series
- less stable estimates
- reduced estimation efficiency

The simulated alternative thinning process showed several extreme spikes, indicating that parameter estimates may become less consistent compared to the standard binomial thinning mechanism.

2.2 Autoregressive and GARCH Processes

2.2.1 Difference Between Autoregressive and GARCH Processes

Autoregressive (AR) Process

An autoregressive process models dependence in the conditional mean of a time series by relating the current observation to previous observations.

For an AR(1) process:

$$Y_t = \phi Y_{t-1} + \varepsilon_t$$

where:

- ϕ is the autoregressive coefficient
- ε_t is a random error term

For discrete count data, the INAR model serves as the integer-valued equivalent of the classical AR model.

GARCH Process

A GARCH process models dependence in the conditional variance rather than the conditional mean. It is mainly used to capture volatility clustering in time series data.

A GARCH(1,1) model is defined as:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where:

- σ_t^2 is the conditional variance
- ε_t is the error term

GARCH models are commonly applied to continuous-valued financial time series.

2.2.2 Comparison Between AR and GARCH Processes

Feature	Autoregressive Process	GARCH Process
Models	Conditional mean	Conditional variance
Captures	Serial dependence	Volatility clustering
Suitable for	Count or continuous data	Mainly continuous data
Discrete counterpart	INAR models	Limited extensions
Variance	Constant	Time-varying

Appendix I: R code for fitting a GARCH Model

A GARCH(1,1) model was fitted to the simulated series.

```
# FIT GARCH(1,1) MODEL
spec <- ugarchspec(
  variance.model = list(garchOrder = c(1,1)),
  mean.model = list(armaOrder = c(1,0)),
  distribution.model = "norm"
)

fit_garch <- ugarchfit(
  spec = spec,
  data = Y_ts
)
fit_garch
```

2.2.3 Interpretation of the GARCH Model

The fitted sGARCH(1,1) model produced a significant AR(1) coefficient (**ar1 = 0.366**, $p < 0.001$), indicating positive serial dependence in the series. The GARCH coefficient ($\beta_1 = 0.999$) was highly significant, suggesting strong persistence in conditional variance, while the ARCH coefficient ($\alpha_1 \approx 0$) was not significant.

The Ljung–Box and ARCH-LM tests all produced p-values greater than 0.05, indicating that the model adequately captured the remaining serial correlation and volatility effects. However, the very small p-values from the Adjusted Pearson Goodness-of-Fit tests suggest that the GARCH model does not fully fit the discrete count nature of the data.

Overall, although the GARCH model identifies persistence in the series, the INAR(1) model remains more appropriate because it is specifically designed for integer-valued count data.

2.2.4 Conclusion

The simulated INAR(1) process successfully captured count-based autoregressive dependence, and the estimated parameters were reasonably close to the true values, indicating good estimation performance. The ACF and PACF plots confirmed the expected first-order dependence structure. While alternative thinning affected the series behaviour, the binomial thinning operator remained more suitable for the classical INAR(1) framework. Although the GARCH model detected persistence in the data, the INAR(1) model provided a better fit because it is specifically designed for discrete count time series.